

EXAM 7 – SPRING 2013

7. (2 points)

The following cumulative reported loss triangle is to be used in a bootstrap process, by building a generalized linear model (GLM) based on incremental loss.

<u>Cumulative Reported Loss</u>			
<u>Accident Year</u>	<u>12</u>	<u>24</u>	<u>36</u>
2010	\$200,000	225,000	250,000
2011	300,000	325,000	
2012	450,000		

a. (0.75 point)

Construct a log-link triangle that can be used to solve for the GLM parameters.

b. (0.5 point)

Now assume that the 2010 cumulative reported loss value at 24 months is \$170,000 instead of \$225,000 and that all other cumulative values stay the same. Briefly explain the obstacles to setting up and solving a GLM bootstrap model for this revised triangle.

c. (0.75 point)

For the revised triangle described in part b. above, construct a log-link triangle that can be used to solve for the parameters of the GLM and briefly describe how to then modify the GLM's fitted values.

CONTINUED ON NEXT PAGE

Question 7 Sample Answer

Solution 1

a) Incremental losses

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	200,000	25,000	25,000
11	300,000	25,000	
12	450,000		

Log link triangle = $\ln(\text{incr. loss})$

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	12.206	10.127	10.127
11	12.612	10.127	
12	13.017		

b) Incremental losses

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	200,000	-30,000	80,000
11	300,000	25,000	
12	450,000		

The GLM bootstrap model assumes all incremental losses are non-negative. The log link function fails for negative inputs. Furthermore, the sum of the 12-24 columns is negative, so the first method of fixing the triangle ($-\ln(\text{abs}(\text{incr. loss}))$) fails.

c) Subtract the largest negative number from every value in triangle, then take log.

<u>revised incr</u>				<u>log link</u>			
<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	230K	0	110K	10	12.346	0	11.608
11	330K	55K		11	12.707	10.915	
12	480K			12	13.082		

modify the fitted values by adding -30K to each value (i.e. subtract 30,000 from all fitted values)

Solution 2

a) $\ln m = \eta_{w,d}$

convert triangle to incremental triangle and take ln of each incremental.

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
2010	$\ln(200,000) = 12.2061$	10.1266	$\ln(25,000) = 10.1266$
2011	12.6115	10.1266	
2012	13.0170		

- b) There will now be a negative incremental value at 24 months for 2010.

$$\text{ie } 170,000 - 200,000 = -30,000$$

Cannot use negative numbers in the model: $\ln(\text{negative \#})$ does not work.

- c) With b) change, incremental loss triangle loss becomes:

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
'10	200,000	-30,000	80,000
'11	300,000	25,000	
'12	450,000		

Since sum of 24 month column is negative, we will take the largest negative value (ie -30,000) and subtract from all values in the table, then take the ln of those values to create the triangle. Set $\ln 0 = 0$.

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
'10	$\ln(230,000) = 12.3458$	0	$\ln(80,000 - (-30,000)) = 11.6082$
'11	12.7068	10.9151	
'12	13.0815		

Solution 3

- c) Now, incremental value is $170K - 200K = -30K$. Sum of column of incremental values is $-30K + 25K = -5K$

We now adjust all incremental values by adding Ψ , where $\Psi = -(-5K) = 5,000$.

Our incremental value at AY 2010, 24 months is still negative so we adjust it by using the log link function as $-\ln(-(-25,000))$ since $-30K + 5K = -25K$.

Once we run GLM, we must remove Ψ from all of the fitted values so we subtract all fitted values by 5,000.

Log-link triangle:

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
2010	12.231	$-10.127 = -\ln(-(-25,000))$	$\ln(85,000) = 11.350$

2011	12.628	10.309 = ln (30,000)
2012	13.028 = ln (455,000)	

Examiner Comment

Most candidates knew how to construct the log link triangle for part a. Common errors for this part included taking the ln of the cumulative amounts rather than incremental or taking the incremental of the ln of the cumulative amount. For part b, most candidates knew that you can't take the ln of a negative number but very few mentioned the secondary issue of the negative sum of the column. Candidates who only indicated that the incremental value was negative without explaining why that was a problem were given no credit. In part c, the candidate had to recognize that the substitution of the $-\ln(-(-30,000))$ wouldn't work due to the negative sum of the column of incremental values. The most common errors included adjusting a subset of the incremental triangle rather than all values in the triangle, using an incorrect value for adjusting, and not correcting the 36 month incremental value for the adjusted 24 month cumulative value.